SCOOT · Level 3 · Mastery

What comes after SCOOT? Six experiments, reported honestly

Connected cars, learning agents, max-pressure, broken loops, emissions and speed advice: each idea built, tested in SUMO and measured fairly.

smrahmanbd.github.io/scoot-learning-lab

7 ideas, each building on the last

1 How to evaluate a new SCOOT idea

9 s spread of SCOOT's delay across three seeds

2 SCOOT on connected-vehicle data

33.9 s delay with 25% connected cars (loops: 30.4 s)

3 A learning split optimiser

67.7 s RL agent delay on unseen days (SCOOT 37.8 s)

4 Max-pressure and spill-back-aware SCOOT

224 s max-pressure delay on 120 m links (SCOOT 28.7 s)

5 Detector faults and historic substitution

70% of the fault damage recovered by historic profiles

6 Emissions-aware SCOOT: the price of a stop

~228 g CO_2 per vehicle for K = 0, 4, 20 and 40

7 Predicting stage changes for GLOSA

1.3-2.6 s mean error of predicted end of green, up to 60 s ahead

Three surprises to start

Better data was worse.

SCOOT running on connected-vehicle counts alone, with every car connected, beat SCOOT on its own loops: 26.3 s against 30.4 s of delay. A loop under a crawling queue over-counts; a simple count does not.

The clever agent lost badly.

A reinforcement-learning split optimiser averaged 67.7 s of delay on unseen days; plain SCOOT averaged 37.8 s. A negative result, clearly measured, is a real research finding and a starting line.

Old data rescued a broken network.

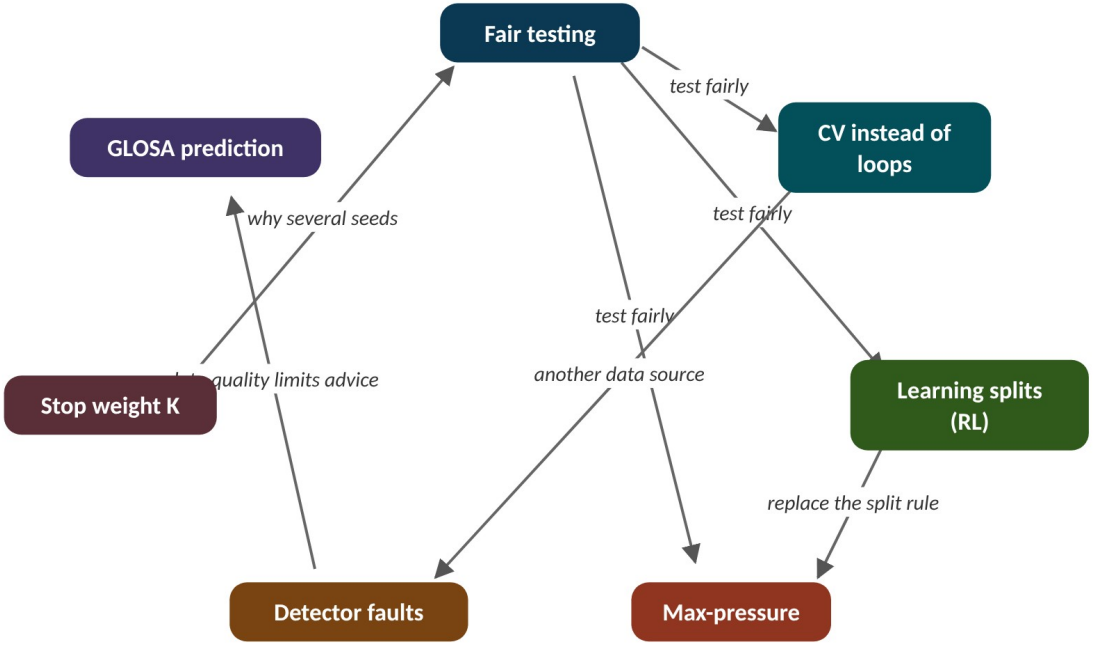
When 13 of 40 detectors failed, trusting their readings more than doubled delay. Swapping in each link's own recorded history recovered about 70% of the loss.

Words you will meet

Baseline	The standard method you compare against: here fixed time, actuation and standard SCOOT.
Seed	The number that sets the 'random dice' of a simulation. Different seeds give different but equally likely days.
Ablation	Switching off one part to see how much it was worth.
Connected vehicle (CV)	A car that reports its position by radio or phone. **Penetration** p is the share of cars that do.
Reinforcement learning (RL)	A program that learns which action to take by trial, error and a score (the reward).

Max-pressure	A controller that serves the movement with the biggest upstream queue minus downstream queue.
Spill-back	A queue so long it backs into, and blocks, the junction behind it.
ASTRID	SCOOT's database of past link data, used here as a substitute for broken detectors.
GLOSA	Green Light Optimal Speed Advisory: telling a driver what speed will reach the light on green.

The map of today's ideas



1 How to evaluate a new SCOOT idea

IN PLAIN WORDS

Testing a new fertiliser? Grow two identical plants in the same soil and light, change only the fertiliser, and repeat with several pairs. One lucky plant proves nothing.

- Change one thing only
- Same demand, same seeds for all
- Several seeds: report the spread
- Count every second, many measures

9 s

spread of SCOOT's delay across three seeds

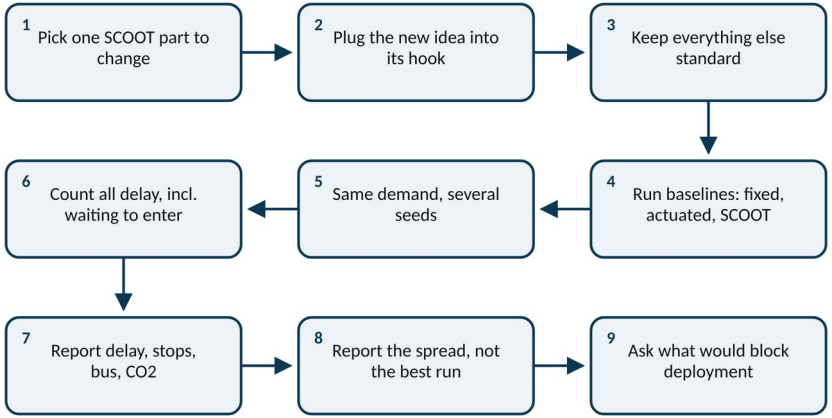
Worked example: Is the difference bigger than the day-to-day spread?

difference = new method − baseline, on the same seed

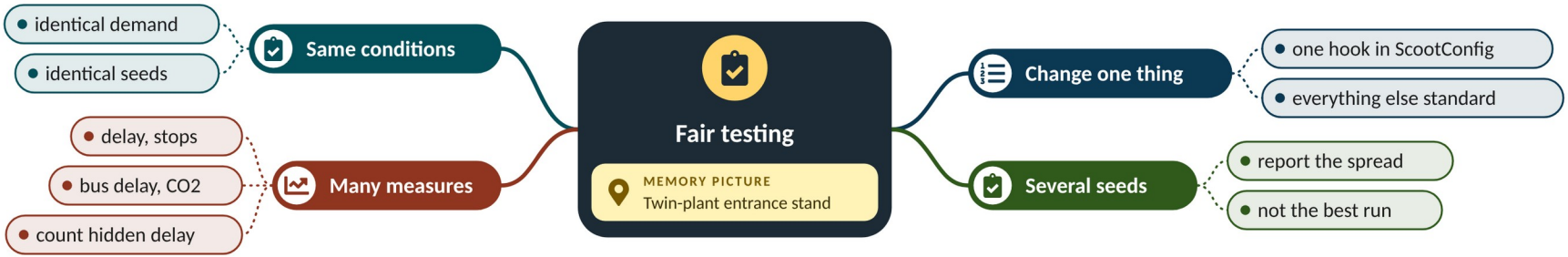
Standard SCOOT on three seeds: 33.1, 38.2, 42.1 s delay. The RL agent on the same seeds: 72.4, 60.8, 69.9 s.

1. SCOOT mean: (33.1 + 38.2 + 42.1) ÷ 3 = 113.4 ÷ 3 = **37.8 s**.
2. SCOOT spread between days: 42.1 − 33.1 = **9.0 s**.
3. Paired differences, RL − SCOOT: 72.4 − 33.1 = 39.3; 60.8 − 38.2 = 22.6; 69.9 − 42.1 = 27.8 s.
4. All three are positive and each is bigger than the 9 s spread.
5. Mean difference: (39.3 + 22.6 + 27.8) ÷ 3 = **29.9 s** worse.

The RL agent is clearly worse, on every day. A 2 s 'improvement' from a single seed would have been inside the 9 s spread and meant nothing.



Remember it: Fair testing



PICTURE THIS · STOP 1: TWIN-PLANT ENTRANCE STAND

At the entrance a judge in a white coat holds two identical potted plants, each with three seeds glued on the pot. She changes the label on ONE pot only, then weighs both on a huge brass scale while a vacuum cleaner under the table refuses to suck any queue off the stage.

TRAP

Do not report your best seed. If the improvement is smaller than the difference between seeds, you have not shown anything yet: run more seeds.

2 SCOOT on connected-vehicle data

IN PLAIN WORDS

A pollster asks 1 in 4 voters and multiplies by 4. Ask enough people and the guess is good; ask 1 in 20 and a few odd answers swing the result wildly.

- Virtual detector: count CVs, divide by p
- Noisy at low penetration
- 25% connected: 33.9 s vs loops 30.4
- 50-100% beat loops: no occupancy inflation

33.9 s

delay with 25% connected cars (loops: 30.4 s)

Worked example: How noisy is one cycle's estimate?

$$\text{LPU}_{\text{est}} = (\text{connected vehicles crossing}) \div p \times \text{LPU per vehicle}$$

$$\text{spread of the estimate} \approx \sqrt{(N\ p\ (1 - p))} \div p$$

20 vehicles cross per cycle. Penetration p = 25%.

1. Expected connected cars: 20 × 0.25 = **5**; estimate 5 ÷ 0.25 = 20. Correct on average.

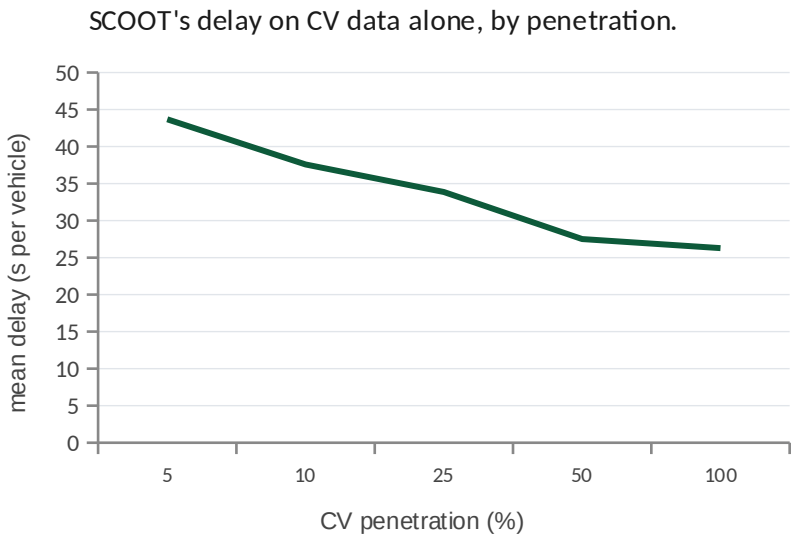
2. Spread of the CV count: √(20 × 0.25 × 0.75) = √3.75 = **1.94** cars.

3. Spread of the estimate: 1.94 ÷ 0.25 = **7.7** vehicles, about 39% of 20.

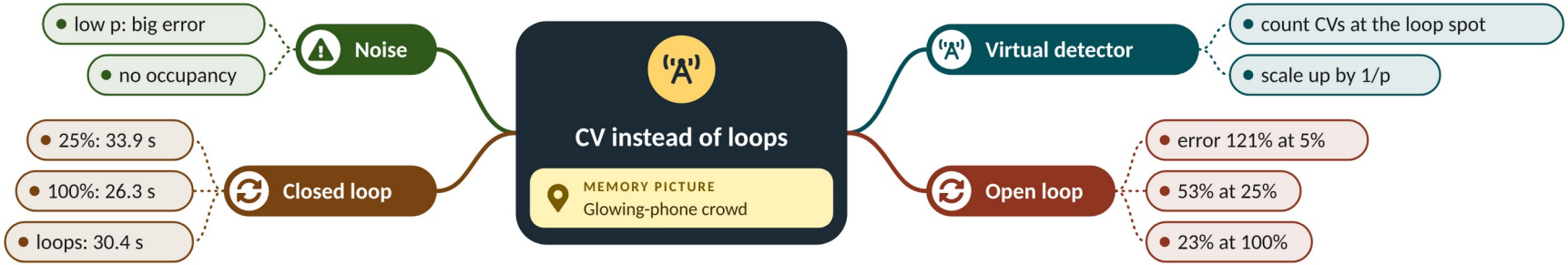
4. At p = 5%: √(20 × 0.05 × 0.95) ÷ 0.05 = 0.97 ÷ 0.05 = **19.5**, about 97%.

5. Chance a cycle sees no CV at 5%: 0.95^{20} = **0.36**.

At 5% a third of cycles see no connected car at all. At 25% one cycle is rough, but smoothing over several cycles brings the error down.



Remember it: CV instead of loops



PICTURE THIS · STOP 2: GLOWING-PHONE CROWD

At the second stand a crowd walks past a chalk line. Only one in four people carries a glowing phone. A bouncer counts the glowing phones and shouts the number times four through a megaphone. When only one person in twenty glows, his guesses jump wildly and he falls off his stool.

TRAP

A big open-loop error does not mean bad control. SCOOT smooths and keeps adjusting, so 53% error per minute still gave delay within 11% of loops. Always test in closed loop too.

3 A learning split optimiser

IN PLAIN WORDS

Teaching a puppy: it tries something, gets a treat or not, and slowly learns. With a clear signal and many tries it learns tricks. With confusing treats and only ten lessons, it just gets excited and knocks things over.

- Learn only the split choice
- Action: −4, 0 or +4 seconds
- Reward: minus the queue change
- First agent lost: 67.7 vs 37.8 s

67.7 s

RL agent delay on unseen days (SCOOT 37.8 s)

Worked example: One Q-learning update

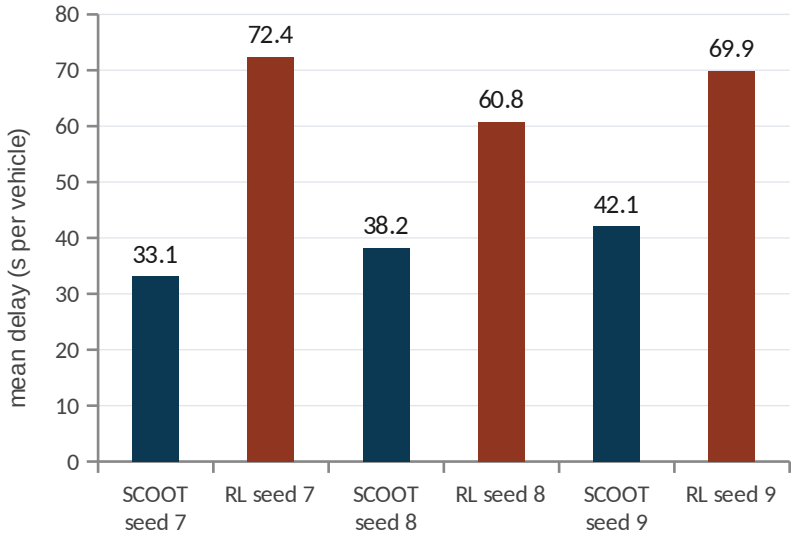
$$Q(s,a) \leftarrow Q(s,a) + \alpha\,[\, r + \gamma \max_{a'} Q(s',a') - Q(s,a)\,]$$

Q(s, +4) = −2.0. The agent takes +4, the queue grows by 1 LPU, so r = −1. In the next state the best Q value is −3.0. α = 0.5, γ = 0.9.

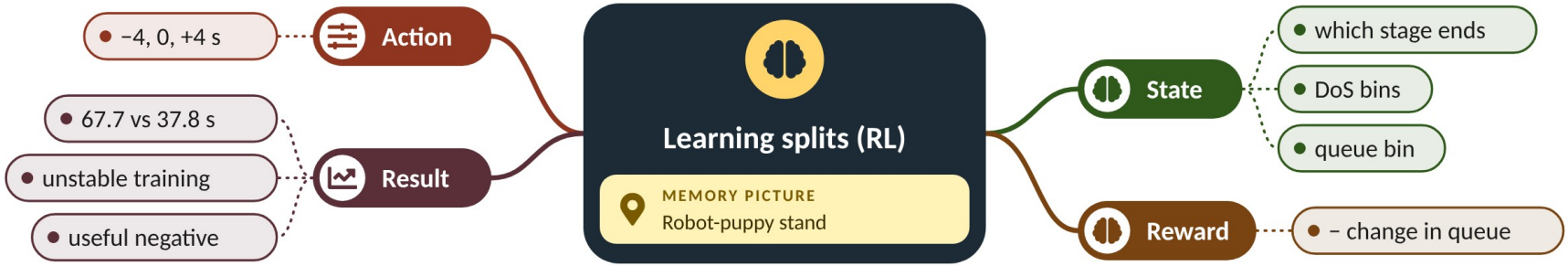
1. Target: r + γ × best next = −1 + 0.9 × (−3.0) = −1 − 2.7 = **−3.7**.
2. Error: target − old = −3.7 − (−2.0) = **−1.7**.
3. Update: −2.0 + 0.5 × (−1.7) = −2.0 − 0.85 = **−2.85**.

The action +4 now looks worse in that state (−2.85), so the agent will try it less. Thousands of such updates are needed; our training had far too few.

Delay on three unseen seeds: SCOOT and the RL agent.



Remember it: Learning splits (RL)



PICTURE THIS · STOP 3: ROBOT-PUPPY STAND

A clockwork robot puppy has three bones labelled −4, 0 and +4. It grabs a bone, gets a treat or a buzz, and after only ten lessons it is spinning in circles, knocking over a sign reading 67.7, while an old sheepdog calmly herds traffic under a sign reading 37.8.

TRAP

A learning agent is not automatically better than a rule. Without enough state information, a reliable reward and enough training, it can be much worse, and its failures are harder to explain.

4 Max-pressure and spill-back-aware SCOOT

IN PLAIN WORDS

A doorman letting people into a room looks at the queue outside. A smarter one also checks whether the room is full: if it is, letting more in only jams the doorway.

- Max-pressure: upstream minus downstream
- No cycle, so no green waves
- Exit-aware SCOOT: discount blocked exits
- Here SCOOT already kept exits clear

224 s

max-pressure delay on 120 m links (SCOOT 28.7 s)

Worked example: Should stage A get more green?

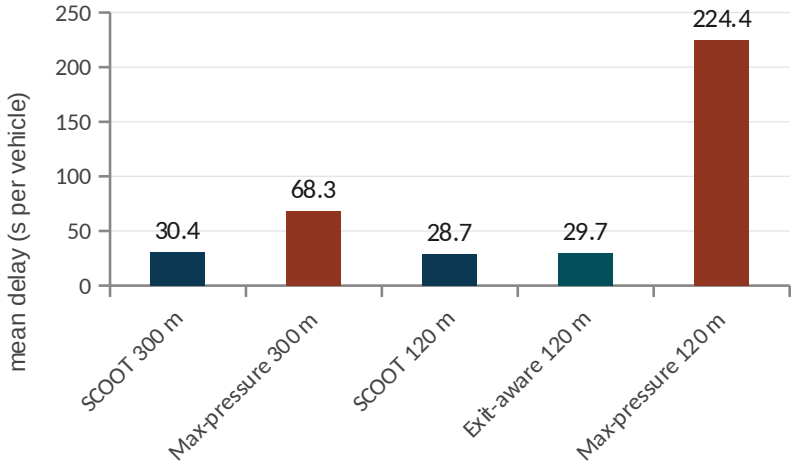
$$\text{blocking} = \max(0, (q_{\text{down}} \div \text{storage} - 0.3) \div 0.7)$$
$$\text{score} = \text{DoS} \div (1 + \lambda \times \text{blocking}), \quad \lambda = 2$$

DoS stage A = 0.95, stage B = 0.80. A's exit link stores 40 vehicles and holds 26. B's exit is empty. λ = 2.

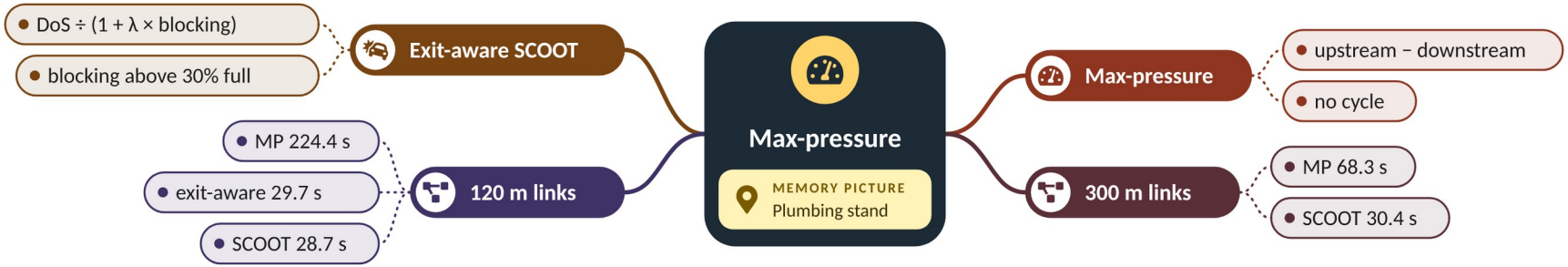
1. A's exit fill: 26 ÷ 40 = **0.65**.
2. Blocking A: (0.65 − 0.3) ÷ 0.7 = 0.35 ÷ 0.7 = **0.5**. Blocking B: **0**.
3. Score A: 0.95 ÷ (1 + 2 × 0.5) = 0.95 ÷ 2 = **0.475**.
4. Score B: 0.80 ÷ (1 + 0) = **0.80**.
5. Plain DoS says A (0.95 > 0.80); exit-aware says **B** (0.80 > 0.475).

Extra green for A would mostly push cars into a filling exit, so the exit-aware rule gives the time to B.

Mean delay: SCOOT, max-pressure and exit-aware SCOOT.



Remember it: Max-pressure



PICTURE THIS · STOP 4: PLUMBING STAND

A wall of pipes with pressure gauges. A plumber opens whichever valve shows the biggest pressure difference, so water squirts in random bursts and never flows smoothly. On the short pipes (120 cm) they burst, soaking a sign reading 224. Next door, a careful plumber divides each gauge by a 'fullness' dial before turning a valve.

TRAP

Max-pressure has strong theory on paper, but it ignores progression and cycle structure. On a coordinated arterial, losing green waves cost far more than its exit awareness gained.

5 Detector faults and historic substitution

IN PLAIN WORDS

If your bathroom scale breaks and says 300 kg every morning, you should not start a crash diet. Better: notice it is impossible and use last month's weight until it is fixed.

- Stuck on: a fake permanent queue
- Stuck off: green shrinks to minimum
- Flag implausible loops, use history
- 69.6 s back to 42.4 s (healthy 30.4)

70%

of the fault damage recovered by historic profiles

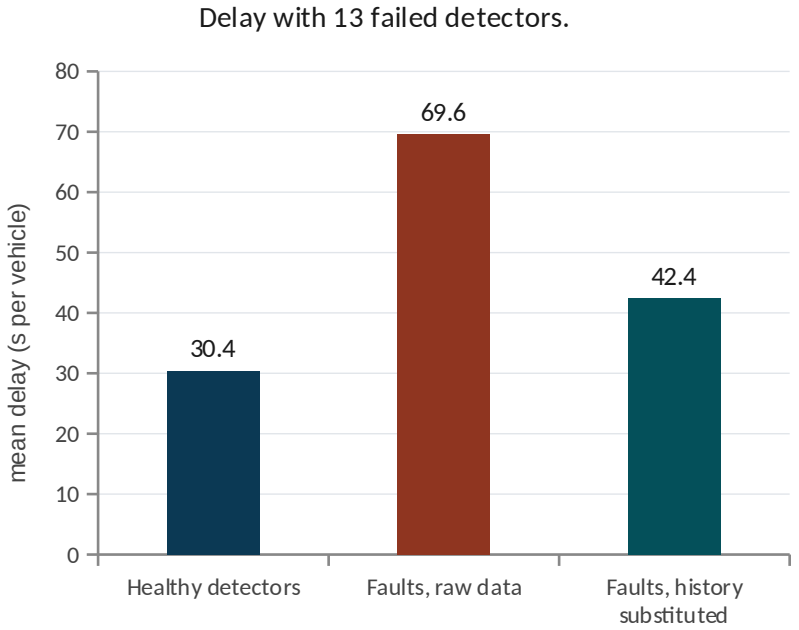
Worked example: How much did history recover?

recovered share = (faulty − substituted) ÷ (faulty − healthy)

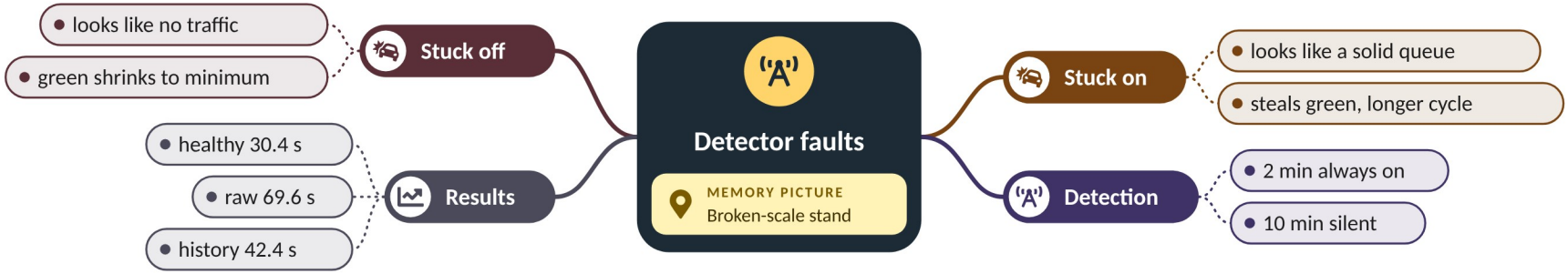
Healthy 30.44 s, faults with raw data 69.62 s, faults with historic substitution 42.44 s.

1. Loss from faults: 69.62 − 30.44 = **39.18 s**.
2. Gained back by substitution: 69.62 − 42.44 = **27.18 s**.
3. Recovered share: 27.18 ÷ 39.18 = **0.69**, about 70%.
4. Still missing: 42.44 − 30.44 = **12.0 s** per vehicle.

History recovers most, not all, of the damage. Better substitutes (neighbouring links, connected vehicles) could close the remaining 12 s.



Remember it: Detector faults



PICTURE THIS · STOP 5: BROKEN-SCALE STAND

A row of 40 bathroom scales; 13 are broken. Some flash 999 forever, others show zero while an elephant stands on them. A librarian with a dusty photo album walks along, covering each broken scale with an old photo of what it used to read. A banner shows 70% in big letters.

TRAP

A stuck-on detector is not 'more data'. It is the most harmful fault, because the model believes there is a permanent queue and reshapes the whole region's timings around it.

6 Emissions-aware SCOOT: the price of a stop

IN PLAIN WORDS

Choosing a route home: you can minimise minutes, or also count each traffic light stop as 'costing' some minutes because stopping is annoying and wasteful. The bigger that cost, the more you prefer a smooth, slightly longer route.

- PI = delay + K × stops
- Tested K = 0, 4, 12, 20, 40
- Most K: ~30 s and ~228 g CO2
- K = 12 outlier: path dependence

~228 g

CO2 per vehicle for K = 0, 4, 20 and 40

Worked example: Which offset wins as K changes?

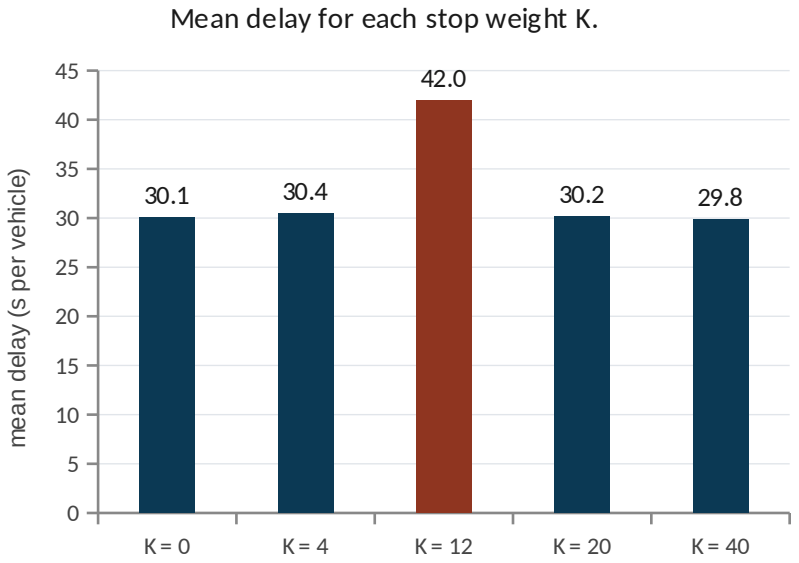
$$PI = delay + K \times stops$$

$$\text{break-even } K = (delay_B - delay_A) \div (stops_A - stops_B)$$

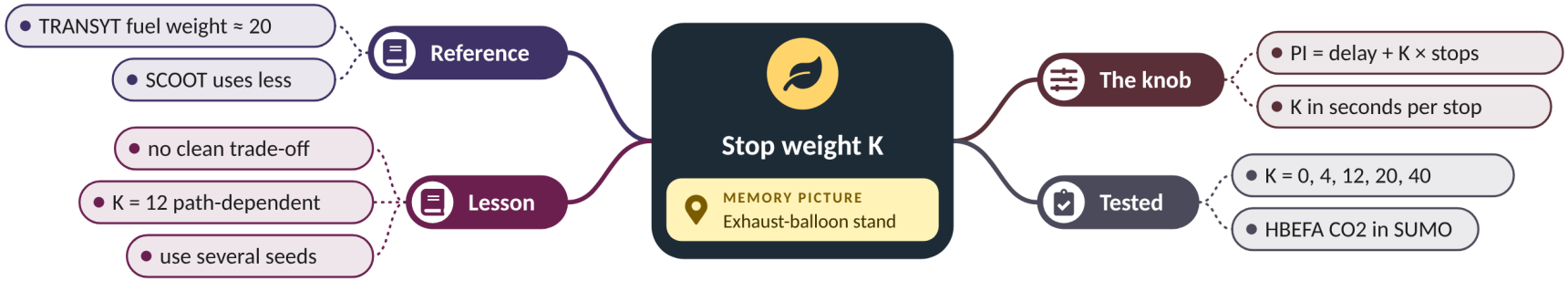
Offset A: delay 300 veh·s, 8 stops per cycle. Offset B: delay 320 veh·s, 5 stops.

1. K = 4: $PI_A = 300 + 4 \times 8 =$ **332**; $PI_B = 320 + 4 \times 5 =$ **340**. A wins.
2. K = 20: $PI_A = 300 + 160 =$ **460**; $PI_B = 320 + 100 =$ **420**. B wins.
3. Break-even: $(320 - 300) \div (8 - 5) = 20 \div 3 =$ **6.7 s per stop**.

Below K ≈ 6.7 SCOOT picks the lower-delay offset; above it, the smoother one. K changes decisions, but not always the final results.



Remember it: Stop weight K



PICTURE THIS · STOP 6: EXHAUST-BALLOON STAND

A toy car drives a little track with a dial labelled K. Every time it stops, a balloon inflates on its exhaust. At K = 0, 4, 20 and 40 the balloons are all the same size, about 228 grams. At K = 12 the car takes a wrong turn at the very first corner and a giant balloon pops.

TRAP

Do not conclude 'K = 12 is bad' from one run. When one setting is worse on every measure and its neighbours are fine, suspect path dependence and run more seeds.

7 Predicting stage changes for GLOSA

IN PLAIN WORDS

A bus timetable that a driver adjusts a little at every stop. Ask 'when does it reach my stop?' a minute ahead and the answer may slip by a couple of minutes; ask when it is one stop away and it is nearly exact, unless something sudden happens.

- GLOSA needs the end of green
- Predict from SCOOT's own schedule
- Mean error 1.3-2.6 s, 90% under 6 s
- Advise with a confidence band

1.3-2.6 s

mean error of predicted end of green, up to 60 s ahead

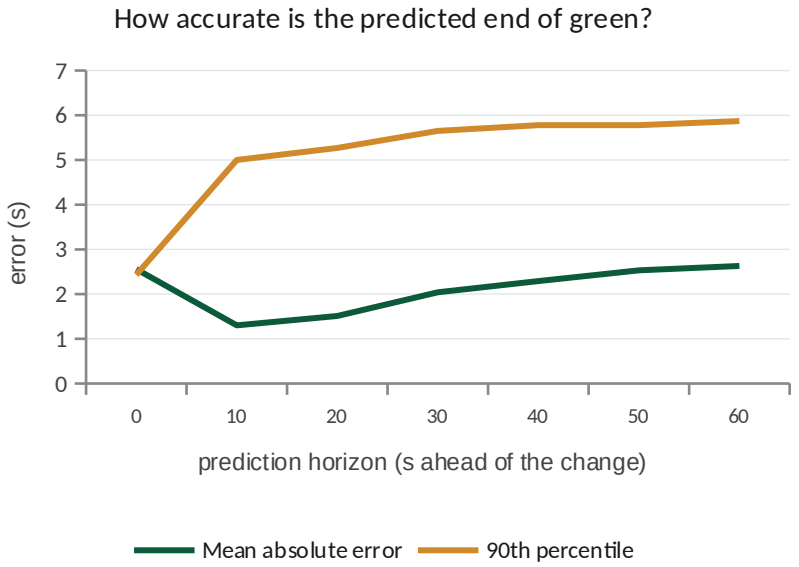
Worked example: Can this driver make the green?

advised speed = distance ÷ time until green
safety margin = predicted time left − time needed

Green predicted to end in 15 s. Car 180 m away, speed limit 13.9 m/s (50 km/h). At this horizon: MAE about 1.5 s, 90th percentile about 5 s.

1. Time needed at the limit: 180 ÷ 13.9 = **12.9 s**.
2. Margin: 15 − 12.9 = **2.1 s**.
3. Compare with MAE 1.5 s: on an average day, the driver makes it.
4. Compare with 90th percentile 5 s: 2.1 s is below 5 s, so on a bad day the green ends first.
5. Advice: do not tell this driver to hurry; plan to stop or advise for the next green.

Average accuracy is good, but advice must respect the worst cases. That is why a confidence band is better than a single predicted time.



Remember it: GLOSA prediction



PICTURE THIS · STOP 7: FORTUNE-TELLER TENT

In the last tent a fortune teller gazes into a crystal ball shaped like a traffic light and announces 'green ends in 15 seconds!'. She is usually off by only one or two seconds, but a bus keeps barging into the tent at the last moment and she yells 'six!' to cover herself.

TRAP

A small average error is not enough for safe advice. Drivers act on single predictions, so the tail (90th percentile, about 5-6 s) decides whether advice is trustworthy.

Six research studies at a glance

Study	What changed	Baseline result	New result	Verdict
CV data	data source	loops 30.4 s	25%: 33.9 s; 100%: 26.3 s	promising
RL splits	split chooser	SCOOT 37.8 s	67.7 s	not yet
Max-pressure	whole controller	SCOOT 30.4 s	68.3 s (300 m)	worse here
Exit-aware SCOOT	split score	SCOOT 28.7 s (120 m)	29.7 s	needs harder test
Detector faults	fault handling	raw faults 69.6 s	history 42.4 s	recovers ~70%
Stop weight K	PI weight	K = 4: 30.4 s, 227 g	no clean trade-off	use more seeds
GLOSA	prediction	n/a	MAE 1.3-2.6 s	usable with a band

Things beginners mix up

This…	…is not this	The fact that tells them apart
Open-loop error	**Closed-loop performance**	53% error per minute at 25% CV still gave delay within 11% of loops.
Best seed	**The result**	Report the spread over seeds; SCOOT alone varied by 9 s.
Max-pressure	**Exit-aware SCOOT**	Max-pressure drops the cycle; exit-aware SCOOT only rescales the split score.
Stuck on	**Stuck off**	Stuck on fakes a queue and steals green; stuck off hides traffic and starves it.
Mean error	**Worst-case error**	GLOSA advice must respect the 90th percentile (~5-6 s), not just MAE.

Check yourself (1 of 2)

20 vehicles per cycle, p = 25%. What is the spread of the CV-based estimate?

A. 1.9 vehicles

B. 5 vehicles

C. 7.7 vehicles

D. 19.5 vehicles

Why did pure max-pressure do badly on 300 m links?

A. It ignores the upstream queue

B. It has no cycle, so it loses progression (green waves)

C. Its detectors failed

D. It used CV data

Answers are in the speaker notes and in the study guide's answer key.

Check yourself (2 of 2)

Offset A: delay 300, 8 stops; B: delay 320, 5 stops. Which wins at K = 20?

A. A

B. B

C. They tie

D. Cannot tell

Exit-aware SCOOT changed no decisions on 300 m links. Why, and what test would be better?

A. Bug; rerun

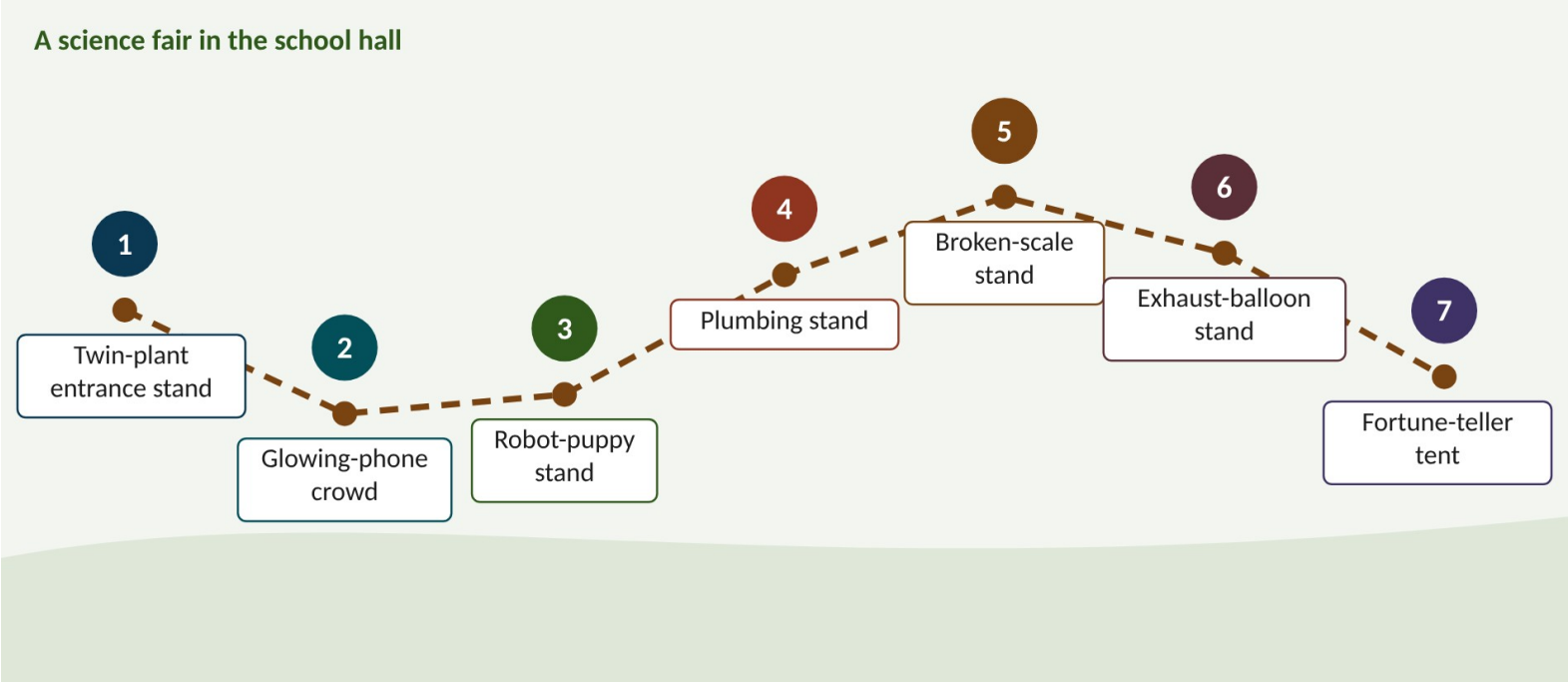
B. No exit ever filled; test a grid of short links with spill-back

C. λ was too large; set λ = 0

D. CV data was missing

Answers are in the speaker notes and in the study guide's answer key.

Walk the memory route



1 Twin-plant entrance stand = Fair testing · 2 Glowing-phone crowd = CV instead of loops · 3 Robot-puppy stand = Learning splits (RL) · 4 Plumbing stand = Max-pressure · 5 Broken-scale stand = Detector faults · 6 Exhaust-balloon stand = Stop weight K · 7 Fortune-teller tent = GLOSA prediction

Numbers to remember

› SCOOT over seeds 7/8/9: 33.1 / 38.2 / 42.1 s (mean 37.8)

› CV delay: 43.7 / 37.6 / 33.9 / 27.5 / 26.3 s at 5-100%; loops 30.4

› RL agent 67.7 s vs SCOOT 37.8 s

› Max-pressure 68.3 s (300 m), 224.4 s (120 m)

› Faults: 30.4 / 69.6 / 42.4 s; 13 of 40 failed; ~70% recovered

› K = 0, 4, 20, 40: ~30 s, 227-230 g; K = 12: 42 s, 244 g

› GLOSA MAE 1.3-2.6 s; p90 under 6 s

Keep going

Study guide: SCOOT_Mastery_Study_Guide.pdf · Interactive labs and SUMO code: smrahmanbd.github.io/scoot-learning-lab